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MILITARY SPECIFICATION

CLEANER, VAPOR PRESSURE, SPRAY RINSE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable combination vapor- and high-pressure rinse, gasoline-engine-operated cleaner.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

P-C-437	Cleaning Compound, High Pressure (Steam) Cleaner
TT-E-489	Enamel, Alkyd, Gloss (For Exterior And Interior Surfaces)
VV-G-101	Gasoline; Motor, United States Government
ZZ-H-541	Hose, Rubber, Steam
ZZ-H-601	Hose, And Hose Assemblies, Rubber (Wrapped Or Braided) Water Service
PPP-C-650	Crates, Wood, Open And Covered

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion On
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 And JP-5
MIL-P-6889	Primer; Zinc - Chromate, For Aircraft Use
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For

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in accordance with NFPA 70, SECTION 422-3. [illegible]

MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-P-10062	Packaging And Packing Of Engines, Spare Or Installed (Other Than Aircraft)
MIL-E-11275	Engines, Gasoline, Industrial Type, General Specification For
MIL-I-11683	Interference Suppression, Radio, Requirements For Engine Generators And Miscellaneous Engines
MIL-I-16165	Interference Shielding, Engine Electrical Systems

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Blocking, Bracing And Waterproofing With Appropriate Test Methods

PUBLICATIONS

U. S. Air Force Bulletin

F49	Codes For Ground Equipment Finishing Schemes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards For Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

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NATIONAL BOARD OF FIRE UNDERWRITERS'

Pamphlet 70	National Electrical Code (Electrical Wiring And Apparatus)

(Copies of the NBFU pamphlet may be obtained upon application to the National Fire Protection Association, 60 Batterymarch Street, Boston MA 02110.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago IL 60606.)

AMERICAN STANDARDS ASSOCIATION

Standard C-50	Rotating Electrical Machinery
Z24.3-1944	Sound Level Meters

(Copies of ASA standards may be obtained upon application to the American Standards Association, 70 East 45th Street, New York NY 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The cleaner shall consist of the following major components:

	Description	See Requirement
a.	Vapor generator	3.8
b.	Heating coils	3.9
c.	Burner	3.10
d.	Fuel tank	3.11
e.	Pump	3.12
f.	Gasoline engine	3.13
g.	Generator	3.14
h.	Control panel	3.15
i.	Concentrate solution tank	3.16
j.	Compound metering device	3.17
k.	Pressure control switch	3.18
l.	Delivery equipment	3.19

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* 3.2.1 The cleaner shall be mounted on a four-wheel trailer with industrial-type pneumatic tires and wheels. The trailer shall be provided with a folding towbar and a towing lunette.

* 3.3 Selection of specifications and standards. Except as specified in 3.3.1 and 3.3.2 below, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exception specified in 3.3.1, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Protective treatment. When materials that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage are used in the construction of the cleaner, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.4.2.1 Dissimilar metals. Wherever possible, contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium alloys shall be avoided. Wherever such contacts are unavoidable, they shall be insulated in a practicable manner.

3.5 Design. The cleaner shall be designed for portable operation in the cleaning of aircraft, vehicles, and miscellaneous ground equipment. All units and assemblies in the cleaner shall be suitable for such use.

* 3.5.1 When used as a vapor cleaner, the cleaner shall have a capacity of 150 gallons per hour (gph) saturated vapor and cleaning solution. When used as a high-pressure rinse unit, the cleaner shall have a capacity of not less than 275 gph of hot or cold water. The cleaner shall be capable of converting from the saturated vapor operation to the high-pressure, hot-or-cold water rinse and vice versa without interruption of cleaner operation.

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* 3.5.2 The cleaner shall operate at a constant pressure at the coil when used as either a vapor cleaner or a high-pressure rinse unit. The cleaner shall deliver the rated volume at not less than 100 pounds per square inch gage (psig) for the vapor cycle and not less than 150 psig for the hot-or-cold water rinse cycle. The rated pressure of the vapor solution and the rated pressure of the hot rinse shall be indicated on the unit.

3.6 Construction. The cleaner shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Mobility.

* 3.6.1.1 The cleaner shall meet the requirements for type II, group C mobility of MIL-M-8090. Manual parking brakes shall be provided on each rear wheel. The lunette eye, reflectors, and weight distribution shall conform to MIL-M-8090.

3.6.1.1.1 Tread width shall be 49 inches. Two wheels shall be located near the rear of the chassis for equal load distribution.

3.6.2 Instruments and controls. All instruments and gages supplied on the unit shall correspond in appearance. All instruments, terminals, conductors, switches, controls, and outlets shall be plainly identified by engraved plates.

* 3.6.3 Cleaning. The cleaner shall be designed and constructed to use steam cleaning compound conforming to P-C-437 to clean dirt, grease, and oils from aircraft, aircraft parts, accessories, and ground equipment by use of a vaporized spray composed of a mixture of steam, hot water and cleaning compound. It shall be designed to use hot water with or without cleaning compound, and cold water through the high-pressure rinse system.

3.6.4 Inflammable liquid containers. All mechanisms and parts designed and constructed to contain inflammable liquids shall be positioned, located, and ventilated to preclude fires or explosions caused by such liquids contacting heated surface or electrical apparatus.

3.6.5 Solution equipment. All solution tanks and solution handling equipment shall be positioned, located, and baffled to prevent spillage of solution over other parts of the cleaner.

3.6.6 Cabinet. The cabinet shall be designed to resist weather and to prevent water from collecting inside or outside the compartment.

3.6.6.1 The cabinet frame and superstructures shall be fabricated from sheet metal having a minimum thickness of 0.064 inch. The side paneling shall be constructed from sheet metal having a minimum thickness of 0.035 inch. The fuel tank and the cleaning solution tank shall be fabricated from sheet metal having a minimum thickness of 0.047 inch.

3.6.7 Enclosure and guards. The cleaner components, including the engine and the auxiliary water tank, shall be fully enclosed in a metal compartment with the exception of valve handles, relief valves, water inlet connections, and gages. All parts not enclosed shall be protected by substantial guards. All working parts shall be accessible through convenient doors or movable panels.

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3.6.8 Screws, pins, bolts, et cetera. All screws, pins, bolts, et cetera, shall be installed with safety wire, self-locking nuts, cotter pins, or lockwashers. All screw threads shall conform to Handbook H28.

3.6.9 Maintenance.

3.6.9.1 Where practical, the cleaner shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases without the use of special tools. Provisions shall be made to exclude dirt, snow, and ice from working parts. Provision shall be made for ease-of-maintenance by personnel wearing heavy gloves. Any design which renders servicing, adjustment, or replacement repeatedly necessary under field conditions shall not be acceptable.

3.6.9.2 Storage. Storage devices shall be provided within the cabinet enclosure, easily accessible from the outside, for storing the tools and the spare nozzle tips when not in use. The storage devices shall be so arranged that stored items cannot be dislodged during transit.

3.7 Performance. The cleaner shall be capable of the following performances:

a. Discharging vaporized spray composed of a mixture of saturated vapor and cleaning compound in accordance with P-C-437 (weight 60 pounds per cubic foot) at not less than 150 gph and 100 psig.

b. Discharging hot water, with or without the specified cleaning compound, to be used through the high pressure rinse system at not less than 275 gph and 150 psig.

c. Discharging cold water through the high pressure rinse system at not less than 275 gph and 150 psig. The cold water high pressure rinse operation shall be accomplished by turning off the burner.

d. Withstanding an electrical system dielectric strength test at a potential equal to twice the working voltage plus 1,000 volts for a period of one minute.

e. Withstanding storage temperatures ranging from -80° to +160° Fahrenheit (F).

f. Operating at a pressure altitude of 8,000 feet at ambient temperatures between 70° and 90°F.

3.8 Vapor generator. The cleaner shall have a vapor generator of the progressive heating, continuous tubular, coil type containing no drums or extended pressure chambers. The vapor generator shall be designed to have a minimum heat generating capacity of 300,000 British Thermal Units per hour output, at a thermal efficiency of not less than 75 percent when operating under maximum load. The minimum thermal efficiency of 75 percent shall be maintained under all heating conditions specified herein. An automatic mechanical means shall be provided to relieve the pressure in the generator when the recommended generator pressure is exceeded.

3.9 Heating coil. The heating coil shall be of welded steel construction and shall withstand a hydrostatic test pressure of not less than 1,000 psig.

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3.9.1 The heating coil shall be designed and arranged to prevent formation of condensate and excessive carbon deposits on any part of the coil.

3.9.2 Coil pressure gages. Two dial-face-type pressure gages reading from 0 to 300 psi shall be furnished and attached to the intake and outlet sides of the heating coils. The gages shall be properly damped against vibration or dial fluctuation and shall accurately indicate the working pressure or coil restriction.

3.9.3 Connectors and fittings. The heating coil shall be equipped with accessible connectors and fittings to permit the circulation of a coil-cleaning solution without resorting to any auxiliary pumping medium.

3.9.4 Combustion chamber. The combustion chamber shall be adequately insulated to prevent excessive heat loss.

3.9.5 Globe valves. Glove valves and suitable fittings shall be provided on the intake and outlet sides of the heating coil as a means of permitting complete drainage and blowdown or to introduce a solution of muriatic acid containing 2 percent (by weight) of paraformaldehyde as an inhibitor for scale-removing operation.

3.9.5.1 The cleaner shall be designed and constructed to permit the introduction and circulation of the scale-removing acid by the solution circulating system without resorting to any auxiliary pumping medium.

3.9.6 Repair and replacement. The heating coil shall be designed and arranged to permit easy removal for repairs and replacement

3.10 Burner. The burner shall be of the pressure-atomizing type equipped with full automatic ignition and controls for smokeless operation under all conditions, including starting and stopping of the unit.

3.10.1 The burner shall be of the automatically- and manually-controlled type. It shall be capable of burning any fuel between the classification of light diesel fuel of 30 degree API gravity, jet fuel conforming to MIL-T-5624, common gasoline in accordance with VV-G-101, and 130 octane aircraft gasoline without requiring any mechanical change in the burner system other than adjustment of fuel pressure.

3.10.2 Location. The burner shall be located so that the burner nozzle or access openings, where fuel or products of combustion might escape, are a minimum of 34 inches above the ground level to minimize danger of explosion of surrounding fuel vapors. Operation shall be such as to permit soot removal at the heater coil with vapor and solution from the cleaning gun while the machine is in full operation. The burner shall be mounted to facilitate cleaning and replacement. The burner shall be designed and located to preclude the possibility of condensate dripping on the burner components.

3.10.3 Operation. The burner shall be of a design and capacity capable of generating the rated operating pressure within 5 minutes from a cold start when operating as a vapor cleaner delivering 150 gph. The burner shall automatically regulate to maintain the constant operating pressure after the cold starting period. When operating as a rinse unit, the burner shall be capable of raising the water temperature a minimum of 130°F above the supply-water temperature when the unit is delivering 275 gph at rated operating pressure.

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3.10.4 Fuel limit device. The fuel system shall be equipped with a limit device which will completely shut off the fuel to the burner when the fuel pressure is insufficient for complete atomization.

3.10.5 Burner safety control. An automatic safety control shall be provided to shut off the fuel and prevent damage in event of firing the unit when totally dry, when there is water shortage or water failure during operation, or when there is any period of excessive vapor generator temperatures.

3.10.6 During operation, the burner shall be unaffected by vibration, sudden shock, or tilted position.

3.10.7 Ignition for burner system. Ignition of the fuel shall be accomplished by a high-potential electrical spark introduced in the ignition electrodes. The ignition system shall provide a spark of sufficient intensity, capacity, and duration to assure positive ignition of the fuels used during all conditions of operation.

3.10.7.1 Safety controls. Safety controls shall be permanently wired into the ignition circuit for continuous protection. The safety control circuit shall be employed to prevent fuel from entering the burner until positive ignition spark exists. Means shall be provided to eliminate any possible burner flash or explosion caused by a hot combustion chamber, to prevent any possible momentary discharge of fuel until perfect ignition is assured, and to protect against explosion or damage caused by such failures as weak ignition spark, cracked ignition electrodes, carbonized ignition points, electric current failure, or any other factor that may contribute to malfunction of the burner.

3.11 Fuel tank. The fuel-tank capacity shall sustain 4 hours of continuous operation under maximum heat load.

3.11.1 The fuel tank shall be subject to atmospheric pressure only. The tank shall be sealed to prevent any possible seepage, splashing, or leakage of other liquids that might contaminate the fuel in the tank.

3.11.2 Fuel tank accessories. The fuel tank shall be provided with a readily-accessible filling inlet, filling strainer, dirt and water separator, shutoff valve, and drain valve or plug. Suitable strainer screens shall be provided in the fuel line between the fuel tank and the burner.

3.11.3 Fuel tank location. The fuel tank shall be so located as to prevent fuel from dripping or splashing on the engine or electrical wiring.

3.12 Pump. The pump shall be of an effective and dependable design and of a positive-displacement type. It shall be capable of handling either water or cleaning solution at a constant delivery of not less than 275 gph at the design pressure when operating as a rinse unit or not less than 150 gph at the design pressure when operated as a vapor cleaner.

3.12.1 Packless construction. The pump shall be of a packless construction to insure a constant rated capacity under continuous operation.

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3.12.2 Pump valves. All intake and discharge valve parts of the pump, such as discs, balls, and spring seats shall be made of stainless steel and shall be so designed that they can be readily replaced. The valves shall be of brass. The pump material shall be resistant to the cleaning acid solution and the cleaning compound.

* 3.12.3 Pulsation dampening. The pump shall be equipped with a pulsation dampening device or air bell which will not waterlog and which will permit periodical drainage.

3.12.4 Pump drain fitting. A readily accessible drain fitting shall be provided in the pump assembly for draining the oil without having to remove other parts of the cleaner assembly.

3.13 Gasoline engine. The cleaner shall be driven by a 2- or 4-cycle, single-cylinder, aircooled gasoline engine conforming to group 1 of MIL-E-11275.

3.13.1 Brake horsepower. The rated, net, continuous brake horsepower of the engine shall be not less than that required to drive the pump and generator, and all accessories, to their specified design capacity.

3.13.2 Engine fuel tank. The engine fuel tank shall be incorporated as a part of the unit assembly and shall have a capacity sufficient to provide fuel for 4 hours or more at rated load operation. A filler opening shall be provided of adequate size and depth to provide ease in refueling. The fuel tank filler opening shall be located in an accessible position.

3.13.3 Transmission of power. Transmission of power from the engine to driven components shall not be made by belts. If a silent chain drive is provided, the chain must be within the confines of the compartment and shielded by a protective cover.

3.13.4 Engine drain fitting. A readily-accessible drain fitting shall be provided in the sump of the engine for draining the oil without having to remove other parts of the cleaner assembly.

3.14 Generator. An engine-driven, alternating-current generator shall be provided to supply power to operate the safety controls, solenoid valves, and the spark for the burner ignition. The rated capacity of the generator shall be not less than 20 percent above that required to operate the listed components without exceeding the specified temperature limits.

3.14.1 Construction. The design and construction of the generator shall conform to good commercial practice and to ASA Standard C-50. The temperature rise shall not exceed the safe limits for the class of insulation used, but in no case shall the temperature rise exceed 131°F above a 104°F ambient.

3.15 Control panel. The control panel wiring and wiring arrangement shall conform to Pamphlet 70. Extreme care shall be exercised in wiring the control panel and marking connections to instruments, switches, and other panel devices. Wiring shall be neatly done in a manner to facilitate maintenance which may involve removal of devices from the panel.

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3.16 Concentrate solution tank. The concentrate solution tank shall be so located and so constructed that the solution will not splash or drip on any component of the electrical system or engine.

3.16.1 Tank operation. The concentrate solution tank shall be equipped with connections and valves to permit filling the tank with water and for subsequent heating and automatic dissolving of the concentrated cleaning compound. The automatic dissolving shall be sufficiently thorough to preclude any possible settling of compound solids within the concentrate tank when compounds of proper specification and properties are used.

3.16.2 Tank drainage. Readily accessible means for draining the concentrate solution tank shall be provided.

3.16.3 Level indicator. The concentrated solution tank or compound container shall be provided with a graduated level indicator marked at 1-1/2-gallon increments.

* 3.16.4 Measuring container. A measuring container with a capacity for 6-1/2 pounds of cleaning compound conforming to P-C-437 (weight 60 pounds per cubic foot) shall be provided. The container shall be marked in increments of 10 ounces or 5/8 pound in order that a 5-percent concentrated solution may be readily obtainable.

3.16.4.1 Means for rigidly clamping or wedging the measuring container to the solution tank shall be provided. The clamping device shall be capable of being easily opened and closed by hand without the use of tools.

3.17 Compound metering device. The cleaner shall be equipped with a compound metering device which will introduce one part of concentrated compound solution to 11.8 parts of clear water. The metering device shall be adjustable throughout the entire range of the concentrate solution tank: zero to not less than 35 gph. When adjusted to the desired metering rate, the devices shall automatically maintain a constant delivery to within ±5 percent at any and all tank-level conditions. Cutoff valves shall be provided in the concentrate solution system so that it can be isolated when not required.

* 3.17.1 Water supply. Unless otherwise specified, the cleaner shall be equipped to secure water from any of the following sources:

a. Tap pressure system from a minimum of 25 up to a pressure of 12 psig.

b. An auxiliary water tank of not less than 100 gallons shall be provided, mounted integrally with the cleaning machine. The tank shall be provided with a constant level valve. The tank shall be equipped with a cover and suitable baffles to prevent splashing of water. Standard garden hose connections and shutoff valves shall be provided. The connection shall be so designed as to eliminate any possible back-siphon or contamination of the water supply source in event of valve failure. The tank shall be provided with a readily-accessible, low-level, drain fitting.

3.17.2 Suitable shutoff valves shall be provided.

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3.18 Pressure-control switch. A pressure-control switch or circuit breaker shall be installed in the electrical circuit which will automatically break the circuit when excessive pressure is exterted on the delivery hose by sudden closing of the nozzle valve, automatically shutting off fuel to the burner and water input to the coil, without stopping the engines. Reopening of the nozzle valve and relieving the pressure in the delivery hose shall automatically close the electrical circuit, and simultaneously re-establish the normal flow of fuel and water through the cleaner and restart the burner.

* 3.18.1 The setting of the pressure switch or circuit breaker shall be adjustable to any set pressure in the range of 50 to 300 psi.

* 3.19 Delivery equipment. One 5-foot, combination vapor-and-rinse cleaning gun assembly shall be provided. The gun assembly shall include a nozzle shutoff valve, swivel handle, and aircooled grips to protect against heat when handling. Nozzles to be provided for the combination gun shall include one rinse nozzle and three vapor nozzles. The vapor nozzles shall include one round nozzle with hardened tip, one 2-inch flat nozzle, and one 4-inch flat nozzle.

* 3.19.1 Delivery hoses. One set of delivery hoses shall be provided with each cleaner as follows. The hose shall be suitable for either vapor or high-pressure-rinse delivery. The delivery hoses shall have an inside diameter of 1/2 inch and shall be in accordance with ZZ-H-541, except that the 1/2-inch male pipe threaded hose nipples shall be held in place with bolt-type clamps. A pressure relief bypass valve shall be installed in the solution system to avoid subjecting the delivery hoses to excessive pressure.

* 3.19.1.1 Hoses. Each cleaner shall be furnished with one 25-foot hose and one 50-foot hose, with all couplings necessary for simultaneously connecting hoses together, to the cleaning gun, and to the cleaner.

3.19.2 Water supply hose. The water supply hose shall be in accordance with ZZ-H-601, grade I, having an oil- and grease-resisting outer cover. The hose shall be 50-foot, and shall be provided with male and female couplings.

3.19.3 Racks. Suitable racks shall be attached to the cleaner for stowing all hoses and the cleaning gun assembly when not in use.

3.19.4 Container. The contractor shall furnish and install a suitable waterproof container to accommodate a handbook of instructions and a parts catalog.

* 3.20 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

* 3.20.1 Handbook of instructions. The contractor shall furnish a handbook of instructions with each cleaner. The handbook shall contain the necessary operating instructions, illustrations, parts lists, manufacturer's part numbers, assembly and disassembly instructions, lubrication instructions, and safety precautions to insure proper operation and maintenance of the cleaner.

3.21 Radio interference suppression. Except as specified below, the cleaner shall be equipped for the suppression of radio interference in accordance with MIL-I-11683.

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3.21.1 Navy. For Navy procurements, the equipment covered by this specification shall comply with MIL-I-16165 for internal combustion engines and MIL-STD-461 for other devices capable of producing radiated and conducted radio interference.

3.22 Noise level. The noise level of the cleaner shall not exceed the limits specified herein for various frequency bands when measured approximately 5 feet above ground level at a distance of not more than three times the maximum diameter of the unit and also at the operator's position and under all operating conditions. Maximum allowable noise level shall be as follows:

Frequency Band (cycles per second)	Noise Level Limit (db above 0.0002 dynes/cm^2)
20 to 75	108
75 to 150	98
150 to 300	90
300 to 600	83
600 to 1,200	78
1,200 to 2,400	75
2,400 to 4,800	72
4,800 to 10,000	71

3.23 Dimensions. The overall size of the cleaner, excluding undercarriage, shall not exceed 48 inches high by 34 inches wide by 95 inches long.

3.24 Weight. The total, uncrated dry weight of the cleaner, including all components described herein, shall not exceed 1,450 pounds.

3.25 Finishes and protective coatings.

3.25.1 Exterior or exposed surfaces. All exposed parts, including iron and steel screws, nuts, bolts, washers, lockwashers, et cetera, shall be suitably protected against corrosion by permanent and impermeable inhibitors. For large bolts and nuts, painting shall be sufficient. The outside of all castings shall be smooth. The complete unit, including all parts except those for which a painted finish may not be necessary, shall be treated in accordance with Bulletin F49, code F49-4, and shall then be given two coats of enamel in accordance with TT-E-489, color 13538 of FED-STD-595. Cadmium plating shall not be used on any exposed surface.

3.25.2 Interior (inclosed) surfaces. All interior surfaces shall be protected with a suitable corrosion-resistant coating.

3.25.2.1 All steel sheets shall be processed with a commercial rust inhibitor.

3.25.2.2 Magnesium alloy parts. Magnesium base alloy parts shall be treated in accordance with MIL-M-3171.

3.25.2.3 Aluminum alloy parts. Aluminum surfaces not subjected to excessive heat shall be primed in accordance with MIL-A-8625. Exterior aluminum alloy sheets shall be primed in accordance with MIL-P-6889 and given two or more coats of enamel conforming to TT-E-489, color 13538 of FED-STD-595.

3.25.2.4 Piping and fittings. All piping and fittings shall be of corrosion-resistant material or the exterior surfaces shall be treated to resist corrosion.

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3.26 Marking.

3.26.1 Instruction panel. Each cleaner shall have an instruction panel, impervious to oil, grease, or weather, mounted on the inside face of the accessible door of the cleaner, and shall include the following information:

a. Operating and maintenance instructions, including detailed instructions for obtaining the proper percentage of concentrate solution.

b. Instructions for blowdown and scale removing.

c. Electrical diagram.

d. Lubrication diagram.

e. Procedure in the event of operating failure.

f. Instructions for removing scale by use of muriatic acid containing 2 percent (by weight) of paraformaldehyde as an inhibitor.

g. Instructions for connecting the water supply system under the various conditions noted under 3.17.1. It should be emphasized that whenever a potable water supply is used as the water source, the anti-siphonage must be used.

3.26.2 The following information shall be permanently and legibly marked on the cleaner in a conspicuous location:

USE ONLY STEAM CLEANING COMPOUND IN ACCORDANCE WITH P-C-437.

3.26.3 A plate impervious to oil, grease, and weather, and marked with the following information in etched or raised 1/2 inch letters shall be securely attached adjacent to the fuel filler cap:

LIGHT DIESEL OIL, JET FUEL, ALL GASOLINES UP TO AND INCLUDING 130-OCTANE AIRCRAFT TYPE CAN BE USED IN THIS CLEANER.

3.26.4 The following shall be permanently and legibly marked in 1-inch letters on the outside front of the cleaner:

USE LIGHT DIESEL OIL, JET FUEL, AND ALL GASOLINES UP TO AND INCLUDING 130-OCTANE AIRCRAFT TYPE ONLY.

* 3.26.5 All controls shall be permanently and legibly marked as to identity or function.

3.27 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.28 Workmanship.

3.28.1 General. The cleaner, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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3.28.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the cleaner, they shall be held or limited accordingly.

3.28.3 Screw assemblies. Assembly screws and bolts shall be tightened to the prescribed torque values for these assemblies.

3.28.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.28.5 Welding. Where welding, brazing, or soldering is used in the fabrication of this assembly, the process used shall be in accordance with the highest commercial standards and in accordance with the latest applicable codes of the American Welding Society. No welding shall be resorted to as a repair measure, except that small blow holes in castings may be filled when not in critical areas, provided that the responsible Government inspector is fully informed of all pertinent details and approves the repair procedure prior to its accomplishment.

3.28.6 Cleaning. The cleaner shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of cleaners shall be classified as follows:

a.	Quality conformance inspection	See 4.3
b.	Preproduction testing	See 4.6

4.3 Quality conformance inspection. Quality conformance inspection tests shall consist of:

a.	Individual tests	See 4.3.1
b.	Sampling plan and tests	See 4.3.2

* 4.3.1 Individual tests. Each cleaner shall be subjected to the following tests, as described under 4.4:

a.	Examination of product	See 4.5.1
b.	Operational check	See 4.5.2

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c.	Dielectric strength	See 4.5.3
d.	Safety check	See 4.5.4

4.3.2 Sampling plan and tests.

4.2.2.1 Lot. A lot shall consist of cleaners manufactured under essentially the same conditions and submitted for inspection at substantially the same time, and may exceed 100 cleaners.

4.3.2.2 Sampling plan. Three cleaners shall be selected at random from each lot and subjected to the following tests as described under 4.5:

a.	Performance demonstration	See 4.5.5
b.	Thermal efficiency	See 4.5.6
c.	Explosion, except only 10 explosions required	See 4.5.12
d.	Static pressure	See 4.5.13
e.	Safety control	See 4.5.14

4.3.2.2.1 Rejection and retest. When one or more cleaners from a lot fail to meet the specification, acceptance of all cleaners in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigations of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.2.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in cleaners already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions.

4.4.1 Apparatus. Apparatus used in conjunction with testing specified herein shall be, insofar as practicable, of laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.4.2 Fuel consumpiton. Fuel-consumption measurement shall be made by the quantitative method whereby the time required for a selected quantity to be consumed shall be measured by a burette. The quantity selected shall be such that each reading will cover an elapsed time of at least 1 minute. All expressions of quantity shall be recorded on a weight basis, and performance curves plotted covering the entire operational range of the cleaner. Performance curves shall include fuel consumption and thermal efficiency versus liquid flow rate.

4.4.3 Liquid flow rates. Liquid flow rates shall be determined by weighing the liquid flowing over a period of not less than 1 minute or by measuring

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the time to fill or empty a container of known volume. Flow meter readings shall not be used unless the flow meter and use thereof has been specifically approved by the procuring activity.

4.4.4 Temperature. Temperatures shall be measured by means of appropriately located thermocouples or thermometers. Insofar as practical, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit.

4.4.5 Pressure. Insofar as practical, pressure and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Gages shall be of laboratory type having a dial diameter not less than 4 inches.

4.4.6 Preparation for test. Except as specified herein, the cleaner shall be prepared for test by servicing with the fuels and lubricants specified by the manufacturer.

4.4.7 Running-in. The nature and extent of running-in shall be specified by the manufacturer.

4.4.8 Discharge pressures. Discharge pressures referred to herein are those measured at the discharge side of the heating coil.

4.4.9 Initial disassembly inspection. Each cleaner submitted for preproduction tests shall be disassembled to the maximum practical extent and completely inspected, the breakdown weights shall be determined, and measurements which are necessary to determine wear and distortion following this inspection shall be made. The cleaner shall be reassembled and subjected to the remainder of the preproduction tests.

4.4.10 Prior to any testing the cleaner shall be serviced in accordance with the instruction of the manufacturer.

4.4.11 Cleaning compound in accordance with P-C-437 shall be used in any and all tests specified hereinafter. The solution shall be as specified herein.

4.4.12 An ample water supply shall be connected to the water supply inlet on the cleaner.

4.4.13 All valves, the pump, and controls shall be checked to assure proper operation of the cleaner. This includes master switch resetting in the burning of various fuels as specified herein.

4.4.14 Data.

4.4.14.1 Readings. During the testing specified herein, at least the following data shall be recorded at intervals not greater than 30 minutes, except where shorter intervals are a test requirement:

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a. Data

b. Time

c. Ambient temperature __°F.

d. Liquid flow rate (water or solution) in pounds per minute.

e. Liquid temperature (water or solution) at outlet from heater __°F.

f. Liquid temperature (water or solution) entering nozzle __°F.

g. Liquid temperature (feedwater to cleaner) entering unit __°F.

h. Water pressure at heater inlet __ psi.

i. Water pressure at heater outlet __ psi.

j. Pressure drop through heater __ psi.

k. Pressure drop to nozzle __ psi.

l. Cabinet temperature __°F.

m. Heater fuel consumption in pounds per hour.

n. Heater input in BTU per hour.

o. Heater output in BTU per hour.

p. Heater thermal efficiency in percent.

4.4.14.2 Notes shall be placed on the logs such as adjustments, servicing, leaks, vibrations, and any other irregularities and corrective measures taken.

4.5 Test methods.

4.5.1 Examination of product. The cleaner shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.5.2 Operational check. The cleaner shall be subjected to a performance demonstration in accordance with 4.5.5 except that the duration of testing at each condition specified shall be reduced to not less than 10 minutes (see 4.5.4).

4.5.3 Dielectric strength. The electrical system shall be subjected to a dielectric strength test at a potential equal to twice the working voltage plus 1,000 volts for a period of one minute.

4.5.4 Safety check. The cleaner shall be checked during operation for general performance of safety features in accordance with procedures specified herein. This checking may be performed during the testing specified in 4.5.2.

* 4.5.5 Performance demonstration. The cleaner shall be operated under the following conditions. Fifty percent of the performance demonstration shall be conducted with aircraft gasoline Grade 100/130 as the fuel for the burner. The

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balance of the performance demonstration shall be conducted on the other fuels specified herein, with the test time divided proportionally.

a. One hour as a vapor cleaner delivering not less than 150 gallons of cleaning solution per hour at not less than 100 psig.

b. One hour with hot water rinse being delivered at the rate of 275 gph at a temperature of 130° +10°, -0°F above the supply water at not less than 150 psig.

c. One half hour with cold rinse water being delivered at the rate of 275 gph at not less than 150 psig.

d. The cleaner shall attain the specified operating pressure in not more than 5 minutes from a cold start while operating as a vapor cleaner at the specified rates and with feedwater being supplied within the normal temperature range of 40° to 70°F. The burner shall automatically regulate to maintain the rated operating pressure after the cold starting period. The cleaner shall also demonstrate compliance with the performance requirements of Section 3 with respect to:

(1) Thermal efficiency.

(2) Temperature regulation.

(3) Pressure regulation.

(4) Fuel consumption.

(5) Pump connections.

(6) Temperature rise in generator insulation.

4.5.6 Thermal efficiency. The thermal efficiency of the vapor generator shall be checked by the following method: The cleaner shall be operated as a rinse unit at design pressure with maximum burner capacity. Measure the water temperature at the inlet and discharge of the unit to determine the temperature rise. Measure the rate of water discharged from the cleaner. With the temperature rise and water rate, determine the BTU output. Measure the rate of fuel consumption to determine the BTU input. From these values the thermal efficiency may be derived.

4.5.7 Radio interference suppression tests. With the exceptions specified in 4.5.7.1, the production test model shall be tested for radio interference suppression in accordance with MIL-I-11683 by authorized representative of the contracting officer. Upon approval by the contracting officer and provided all requirements of this specification are met, the production test model shall be used as a model for all production units.

4.5.7.1 Navy. Tests for radio-interference suppression shall be conducted on the equipment described hereinbefore, and all testing equipment, instruments, personnel for making the test location which shall be reasonably free from radiated and conducted interference, and other necessary facilities shall be furnished by the contractor. Tests for radio interference will not be required for equipment that is identical physically and electrically to those that have met the requirements of MIL-I-16165 or MIL-STD-461.

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4.5.8 Operation in tilted position. The complete cleaner assembly shall be tilted in different positions at an angle of 15 degrees from the horizontal plane. The cleaner shall be operated for 30 minutes at each condition as specified in 4.4.5 in the various positions. The unit shall be permitted to stand for 4 hours in the tilted position most conducive to fuel leakage. The unit shall then be started and operated for a period of 15 minutes. Observations shall be made to determine that complete lubrication is maintained during all phases of the tilted operation.

4.5.9 General utility. The general utility of the cleaner shall be determined by cleaning surfaces that are contaminated with heavy oil and grease deposits, directing the vapor and water rinse streams into deep channels or crevices packed hard with mud, and the combinations of oils, grease, dirt, and other foreign matter that will collect on the undercarriage of aircraft and vehicles during normal operation and by cleaning corrosion preventive compounds from equipment so treated for long storage periods.

4.5.10 Serviceability. The serviceability of the complete assembly shall be evaluated from the standpoint of ease of operation, servicing, and maintenance. Particular attention shall be directed toward maintenance with a minimum number of common tools, servicings, and operation by personnel wearing heavy gloves and provisions made to exclude dirt, snow, and ice from working parts.

4.5.11 The noise level generated by the cleaner shall be checked using noise level meters conforming to ASA Standard Z24.3. Readings shall be taken at several points at a distance of 10 feet from the perimeter of the test specimen. The noise level reading at any point shall not exceed that specified (see 3.22) under all operating conditions.

4.5.12 Explosion. With the combustion chamber and heat exchanger components at room temperature, a gaseous fuel-air mixture in a ratio of 0.085 to 0.095 shall be introduced into the combustion chamber. The chamber shall be purged with this mixture to the extent of at least 10 times the volume of the chamber. The mixture shall be ignited with the heater ignitor. This procedure shall be repeated 50 times. The cleaner shall exhibit no deleterious effects of the explosions and thereafter shall complete all ensuing tests in a satisfactory manner.

4.5.13 Hydrostatic tests. When subjected to the following tests, the cleaner shall exhibit no deleterious effects as a result of these tests:

4.5.13.1 Pressure solution system. The pressure solution system of the assembled cleaner shall be subjected to a hydrostatic test of not less than 1-1/2 times the maximum working pressures.

4.5.13.2 Heating coil. The heating coil shall be subjected to a hydrostatic test of 1,000 psi.

4.5.14 Safety control. The burner safety controls shall prevent any burner flash or exposion during the following tests while burning gasoline fuel. The safety controls shall also automatically re-establish burner flame within one minute after return to normal operation of the cleaner.

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4.5.14.1 After the cleaner has been in operation for a minimum of one hour to insure normal combustion chamber temperature, malfunction of the cleaner shall be caused by the following methods to determine proper function of the burner safety controls:

4.5.14.1.1 Fuel system interruption. While the cleaner is in full operation, the burner control valve shall be turned to the OFF position for a period of 10 seconds and then the valve shall be turned to the ON position. After the fuel system has been so interrupted, the safety controls shall automatically allow a complete purging of the combustion chamber before the ignition can be restarted.

4.5.14.1.2 Water shortage or failure. While the cleaner is in full operation, the water supply shall be turned off until the temperature control stops the burner. Immediately after the burner has been stopped, the water supply to the cleaner shall be turned on. The cleaner shall then automatically regain normal solution capacity and safe burner operation.

4.5.14.1.3 Shortage or depletion of fuel supply. While the cleaner is in full operation a feed line nut on the fuel pump suction side shall be loosened, thus permitting air to enter the fuel pump suction system simulating fuel shortage. If gravity feed fuel supply system is used, the fuel supply shall be stopped by a valve in the supply line. After the burner has been stopped, thus simulating fuel shortage, the feed line nut shall be tightened and the burner control valve turned off, thus permitting the fuel pump to re-prime. After the burner has been turned to the ON position, the safety controls shall automatically reset and control the burner during the safety cycle.

4.5.14.1.4 Power source failure. While the cleaner is in full operation, the power supply to the cleaner shall be broken for momentary interruptions. After each such interruption, the safety controls shall automatically reset for safety cycle.

4.5.14.1.5 Ignition failure due to faulty components. While the cleaner is in full operation the nozzle control valve shall be shut off, thus causing automatic shutdown of the cleaner. The ignition electrodes shall be grounded and the nozzle control valve opened, thus allowing the engine and pump to restart. Under these conditions, the burner safety controls shall prevent fuel from going to burner, thus simulating a condition of a cracked electrode, burned-out transformer, or carbonized ignition points.

4.5.15 Mobility. The cleaner, with all trailer tanks filled, shall be subjected to the applicable mobility test of MIL-M-8090 (see 3.6.1). Failure to meet these mobility requirements shall be cause for rejection of the cleaner.

4.5.16 Environmental. Environmental tests and endurance tests shall be conducted on separate units so that the time required for testing will be held to a minimum and that the endurance tests may be conducted on equipment not requiring a complete disassembly for cleaning previous to conducting endurance tests.

4.5.16.1 High temperature exposure. The cleaner shall be placed in a test chamber and exposed to high ambient temperatures in accordance with high temperature tests, procedure I of MIL-E-5272. Upon completion of the exposed period, the cleaner shall be removed from the test chamber and examined. There shall be no evidence of softening of thermoplastic materials

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or other damage. Upon completion of the cooling-down period, the cleaner shall be subjected to and satisfactorily pass the performance-demonstration test specified herein.

* 4.5.16.2 Altitude. The cleaner shall be placed in a test chamber or under conditions to maintain a pressure equivalent to 8,000 feet altitude and an ambient temperature between 70° and 90°F. It shall then be subjected to the performance-demonstration test specified herein, except that the operating time at each test condition shall be one half of that specified.

4.5.16.3 Low temperature exposure. The cleaner shall be placed in a test chamber and exposed to an ambient temperature of -80° ±2°F for a period of 24 hours. Upon completion of the exposure period, the cleaner shall be removed from the test chamber. After warming up to normal temperature, the cleaner shall satisfactorily complete a performance-demonstration test as specified herein.

* 4.5.17 Endurance testing. The cleaner shall be sujbected to a 250-hour endurance test as specified in 4.5.17.3. The last 100 hours of the endurance test shall be conducted with aircraft gasoline Grade 100/130 as the fuel for the burner. The balance of the endurance test shall be conducted on the other fuels specified herein. with the test time divided proportionally.

4.5.17.1 Inspection, servicing, and adjustments. Planned stops shall be made at 25-hour intervals during the endurance test for inspection, servicing, and adjustments. Records shall be kept of any and all adjustments that may be of a minor nature. Any major adjustments shall be cause for rejection of cleaners.

4.5.17.2 Test time. Intervals of endurance test time less than one hour terminated by a parts failure shall not be credited to the endurance test. Except where shorter periods are a test requirement endurance test time shall be credited in 1/2 hour intervals.

* 4.5.17.3 Test. The 250-hour endurance test shall be conducted as follows:

a. 125 hours as a vapor cleaner at normal operation of 150 gallons of hot solution per hour at rated heating coil discharge pressure.

b. 125 hours as a hot water rinse at normal operation of 275 gallons water per hour at rated heating coil discharge pressure. This test shall be conducted with the cleaner being operated to simulate actual cleaning of a vehicle or aircraft.

4.5.18 Performance check. Immediately following the endurance test and without intervening reconditioning, the cleaner shall be subjected to the performance demonstration specified herein. The loss in performance shall not exceed 2 percent below specified requirements.

4.5.19 Teardown inspection. Upon completion of the specified testing, the cleaner shall be completely disassembled for inspection.

4.5.19.1 Parts subject to wear or distortion shall be measured or otherwise dimensionally inspected and the measurements shall be compared with similar data compiled prior to the start of the prescribed testing for determination of wear or distortion. All parts shall be required to be in acceptable condition. Wear or distortion which require rework or replacement of the parts affected shall be cause for considering the part affected as having failed to complete the test satisfactorily.

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4.6 Preproduction testing.

4.6.1 Preproduction test samples. The preproduction test samples shall consist of 2 cleaners representative of the production cleaner. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

* 4.6.2 Preproduction test report. A preproduction test report shall be prepared in accordance with MIL-STD-831 and three copies of the report submitted to the procuring activity.

4.6.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.5.

5. PREPARATION FOR DELIVERY

5.1 Level of preservation and packaging.

5.1.1 Level A.

5.1.1.1 Unless otherwise specified, the components of the cleaner other than the gasoline engine shall be preserved and packaged in accordance with method IC of MIL-P-116.

5.1.1.2 Gasoline engine. The gasoline engine shall be preserved and packaged in accordance with the applicable requirements of MIL-P-10062.

5.1.2 Level C. Cleaners shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Levels of packing.

* 5.2.1 Level A. The cleaner preserved and packaged as specified in 5.1.1 shall be packed in a covered crate conforming to PPP-C-650, style B, type 4. The crate shall be closed, strapped, and marked in accordance with appendix to crate specification.

* 5.2.2 Level B. Same as 5.2.1 except the PPP-C-650 crate shall be an open type crate.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Uniform Freight Classification rules or regulations of other common carriers, as applicable to the mode of transportation.

* 5.2.4 Physical protection. All cushioning blocking, bracing and bolting as required shall be in accordance with MIL-STD-1186.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The cleaner covered by this specification is intended for use in washing and cleaning of aircraft engines and accessories, vehicles, and miscellaneous ground equipment by the application of vapor pressure, cleaning compound solution, and water rinses.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Point of inspection for preproduction tests.

c. Selection of applicable levels of preservation and packaging, and packing (see Section 5).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - YD
Air Force - 82

Review Activities:
Army - WC
Navy - YD

Preparing Activity:
Air Force - 82

Project No. 4940-0289

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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